

Project Design Phase-II Data Flow Diagram & User Stories

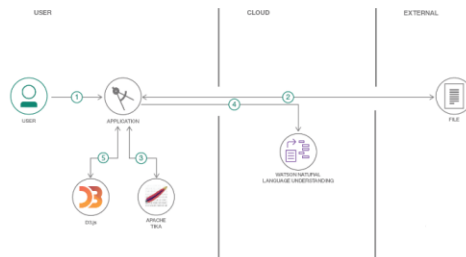
Date	1 JULY 2026
Team ID	SWTID-2026-2662
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

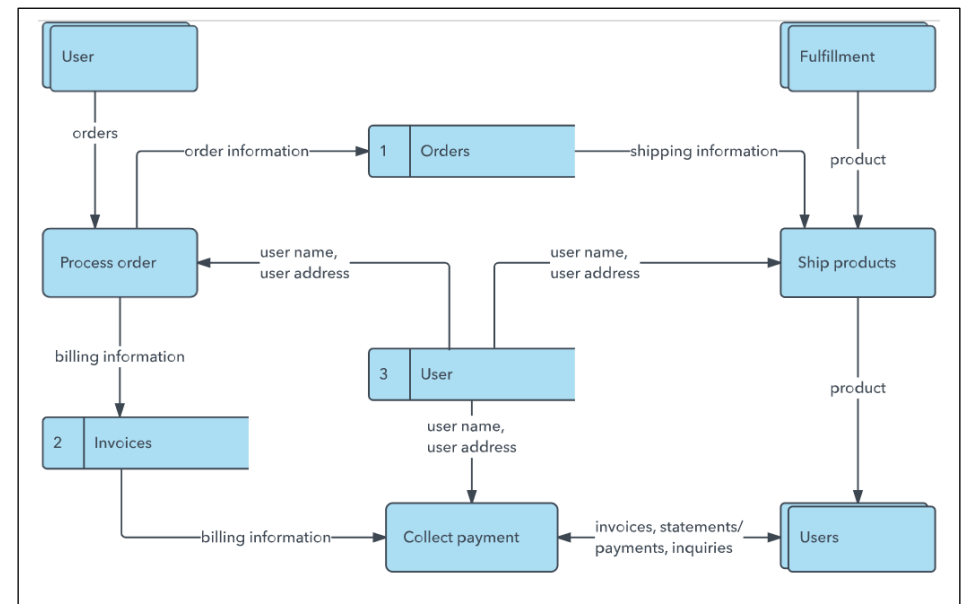
Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Borrower Profile / Admin	User Onboarding & Auth Management	USN-1	Implement secure registration, password hashing (werkzeug or bcrypt), and JWT middleware generation routers in FastAPI.	Emails must be globally unique in SQLite database.Rejects negative income or expense profile parameters.Generates a valid bearer token on secure matching.	High	Release 1.0 (MVP Baseline)
Active Debt Borrower	Core Liability & Debt Logging	USN-2	Develop dynamic data forms to capture loan parameters (OutstandingAmount, EMI, OverdueMonths) and handle CRUD operations via SQLAlchemy.	Integrates relational cascade pathways (delete-orphan) if parent profiles clear.Form validations reject blank creditor entries.	High	Release 1.0 (MVP Baseline)
Financial Analyst Interface	Dynamic Metric Processing Engine	USN-3	Construct backend programmatic math utilities (financial_engine.py) to derive calculated values for DTI metrics and surplus tracking.	DTI evaluates precisely using structural data logic: $\frac{\text{Total EMIs}}{\text{Income}} \times 100$.Triggers custom stress categories automatically.	Medium	Release 1.0 (MVP Baseline)
Distressed Account Holder	AI Settlement Target Generator	USN-4	Build automated calculation prediction tools (settlement_engine.py) to supply localized target range numbers.	Saves computed output variables smoothly down to Settlement_Prediction tables.Ties risk tags to target ratios.	High	Release 1.1 (AI Drop)
Distressed Account Holder	Generative Negotiation Letter Workspace	USN-5	Integrate the google-generativeai SDK into the FastAPI pipeline to compile optimized letters for active creditors based on user financial limits.	Properly builds structured user data parameters directly into the prompt framework.Outputs copy-	High	Release 1.1 (AI Drop)

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
				ready Markdown text layouts.		
System Operator / Admin	Generative Negotiation Letter Workspace	USN-6	Engineer local rule-based text replacement loops to handle unexpected Gemini API credential exceptions or timeout limits.	Smoothly intercepts downstream network connection faults.Renders valid fallback text drafts instantly to preserve the user experience.	Medium	Release 1.1 (AI Drop)
Active Debt Borrower	Strategic Historical Audits	USN-7	Create historical log tracking views to pull, display, and review past AI letters and generated data metrics from previous runs.	Queries rows cleanly from AI_History relational database indexes.Appends accurate system tracking timestamps.	Low	Release 1.2 (Audit Cycle)